

Cryptography and Network Security

Course Code: **CSEG3040**

Instructor: [AMIT GURUNG](#)

Email: amit.gurung@ddn.upes.ac.in



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Cryptography and Network Security

Third Edition
by William Stallings

More details: Refer to the Course plan

***When do you think security
concepts/cryptography first used?***



<https://www.menti.com/ale3m3sdikox>

When where they first used?

- Hieroglyphics (complex writing of ancient Egypt) https://www.youtube.com/shorts/AL4eTiV_XvI



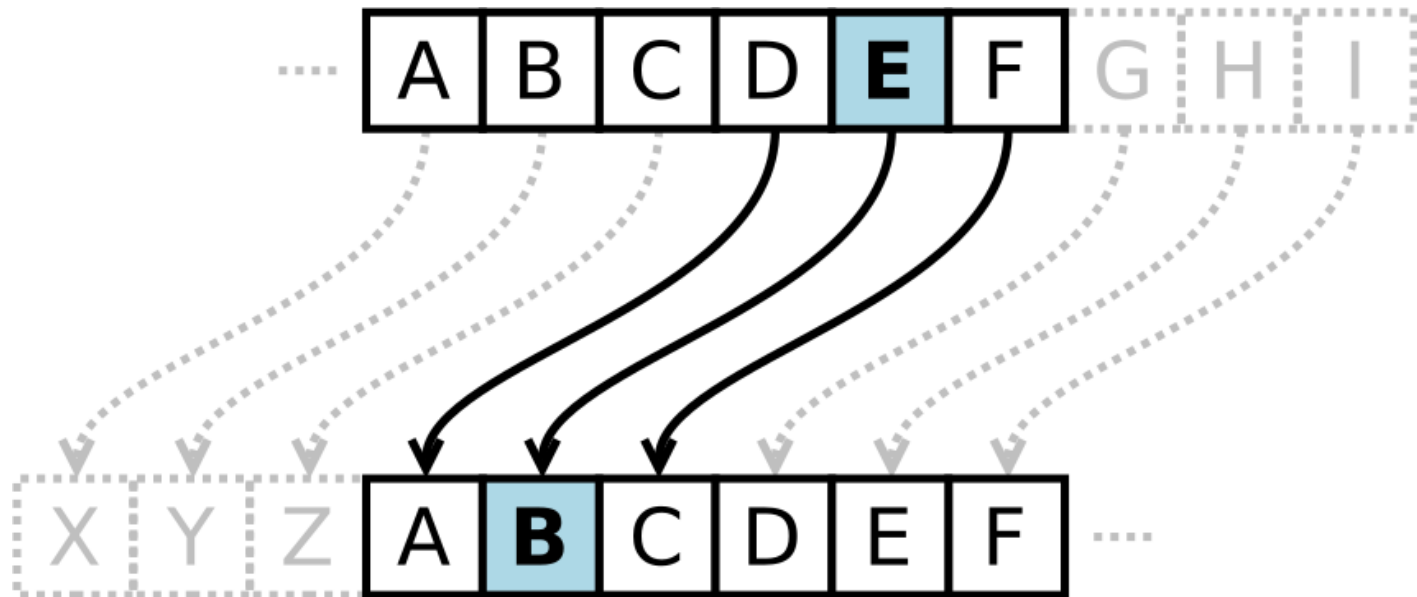
When where they first used?

- Scytale (ancient Greeks, and the Spartans) <https://www.youtube.com/shorts/9CuIgJPV1QE>



When where they first used?

- Julius Caesar, the Roman military general and leader, used the cipher for military communication



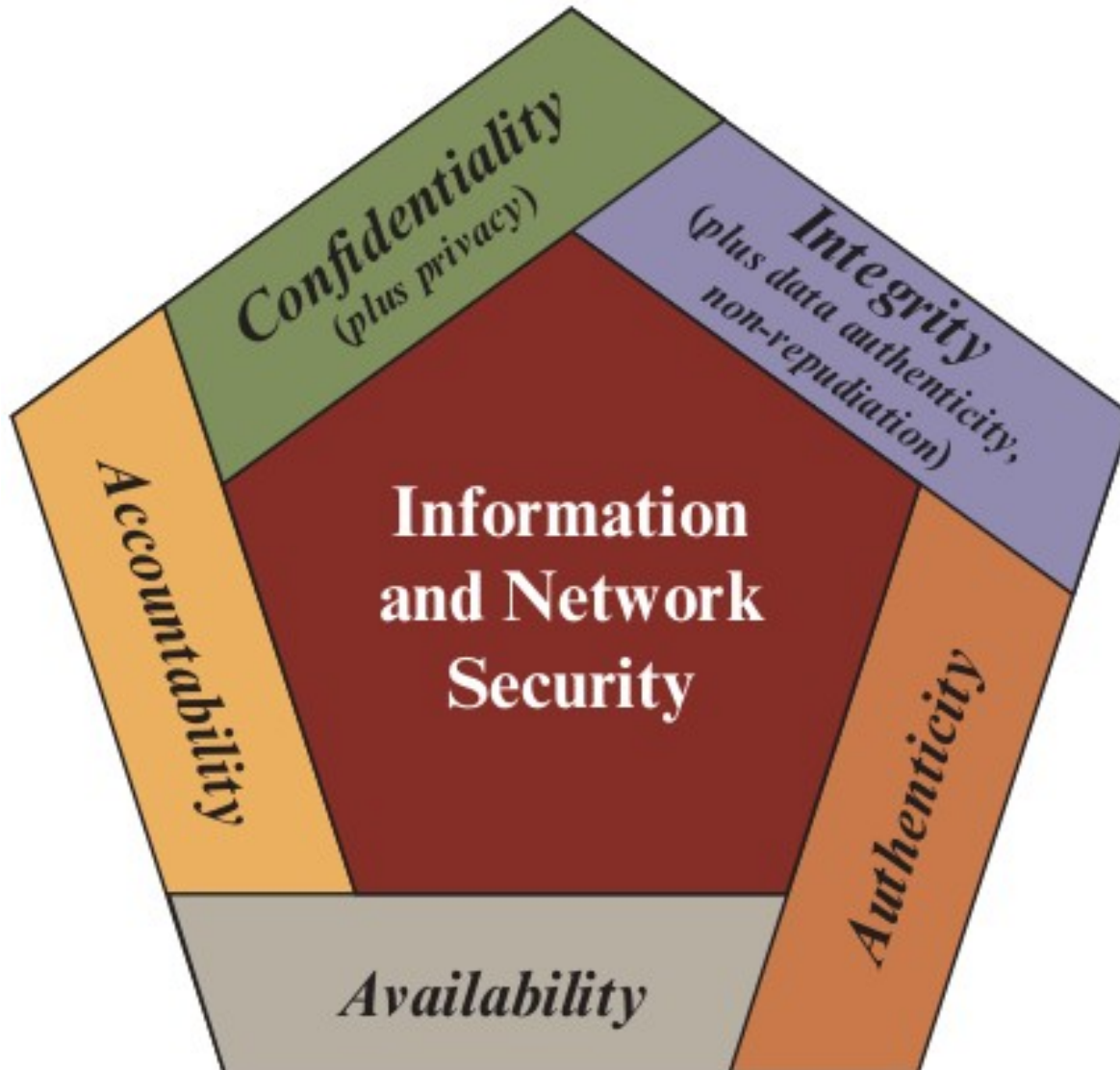
Background

- Information Security
- Tools to protect files and other stored information
- Networks and communications links requires measures to protect data during transmission

Definitions

- **Computer Security** - generic name for the collection of tools designed to protect data and to thwart hackers
- **Network Security** - measures to protect data during their transmission

Key Security Concepts



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Learnings: What did I Learn?

Reflections: What does this make me think about?

Action: How will I use this knowledge?

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Recap: What did I Learn?



Recap: What does this make me think about?



Recap: How will I use this knowledge?

studied knowledge privacy games purposes job
understand build Ill now dive go use today
Examination later password data dont
related topics practical sem knowlegde
study safely real better guess
project clear prior certain future securing
facilitate know secure peculiar will cipher grades
Maybe internet may much Learn life make safe
research cryptography exam life techniques try
keep subject personal subejct communication protect
educational

Some more basic concepts and terminology

Threat, Vulnerability and Attack

Threat

A possible danger that could harm your system.

Example: A hacker.

Example: natural threat (earthquakes).

Threat, Vulnerability and Attack

Vulnerability

A weakness in your system that a threat can exploit.

Example: Unpatched software or a weak password.

Example: tall and weak buildings.

Threat, Vulnerability and Attack

Attack

An actual action where the threat uses the vulnerability.

Example: Hacker uses the bug to steal data.

Example: buildings fallen due to earthquakes.

Services, Mechanisms, Attacks

- Need systematic way to define requirements
- Consider three aspects of information security:
 - security service
 - security mechanism
 - security attack

Security Services (X.800)

- **Authentication** - assurance that the communicating entity is the one claimed
- **Access Control** - prevention of the unauthorized use of a resource
- **Data Confidentiality** –protection of data from unauthorized disclosure
- **Data Integrity** - assurance that data received is as sent by an authorized entity
- **Non-Repudiation** - protection against denial by one of the parties in a communication

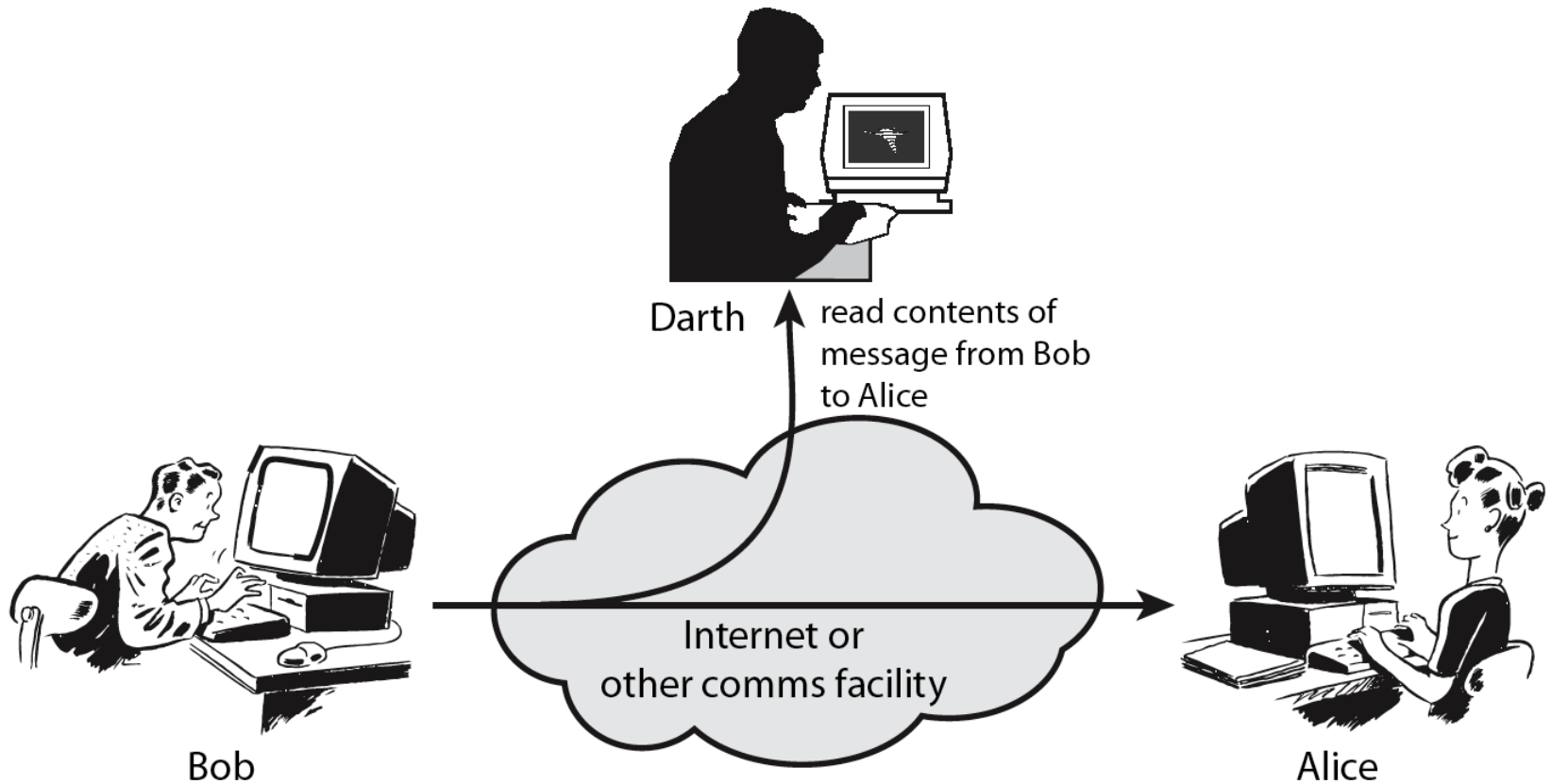
Security Mechanisms (X.800)

- specific security mechanisms:
 - encipherment, digital signatures, access controls, data integrity, authentication exchange, traffic padding, routing control, notarization
- pervasive security mechanisms:
 - trusted functionality, security labels, event detection, security audit trails, security recovery

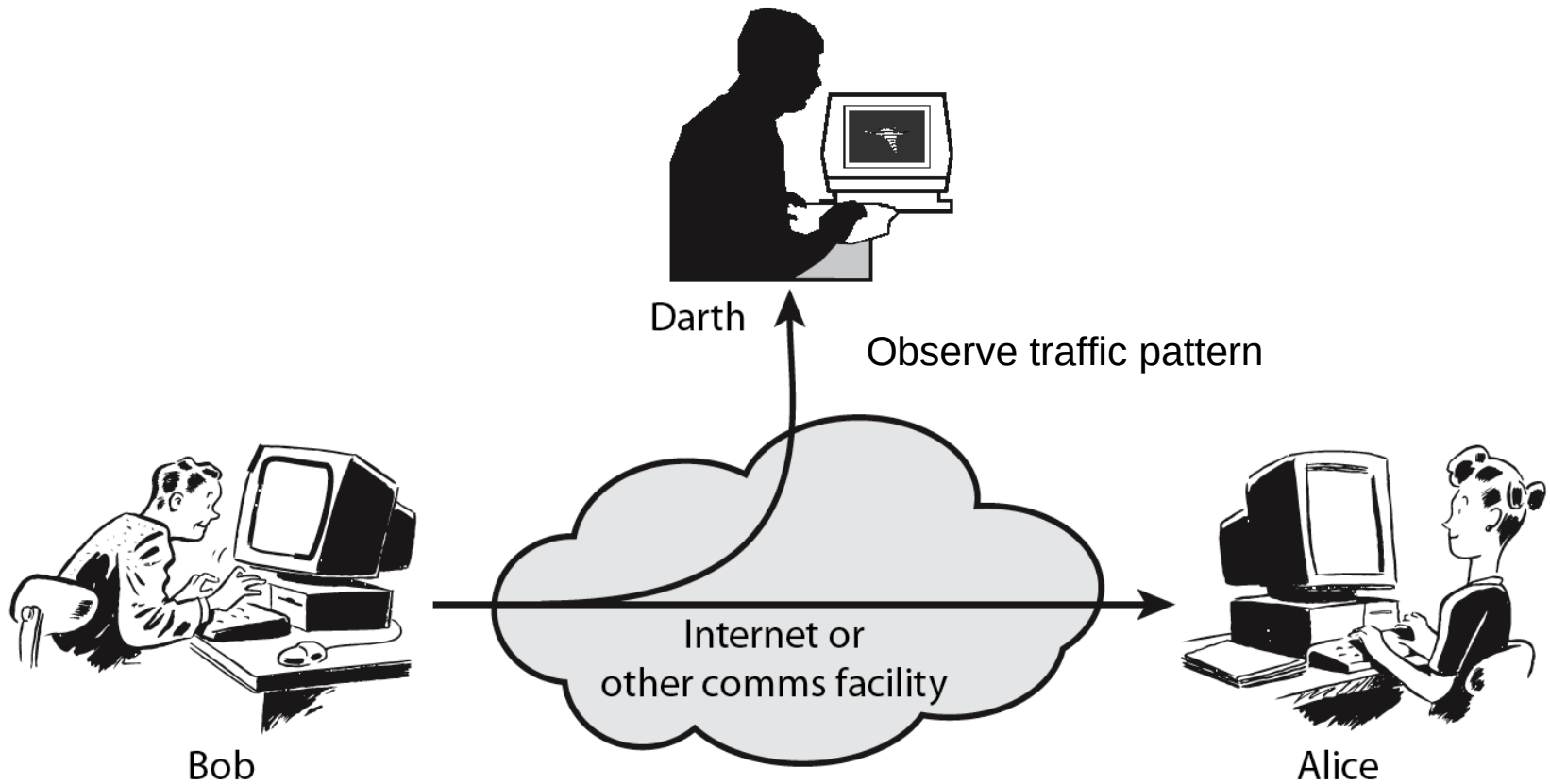
Classify Security Attacks as

- **passive attacks** - eavesdropping on, or monitoring of, transmissions to:
 - Release of message contents, or
 - monitor traffic flows
- **active attacks** – modification of data stream to:
 - masquerade of one entity as some other
 - replay previous messages
 - modify messages in transit
 - denial of service

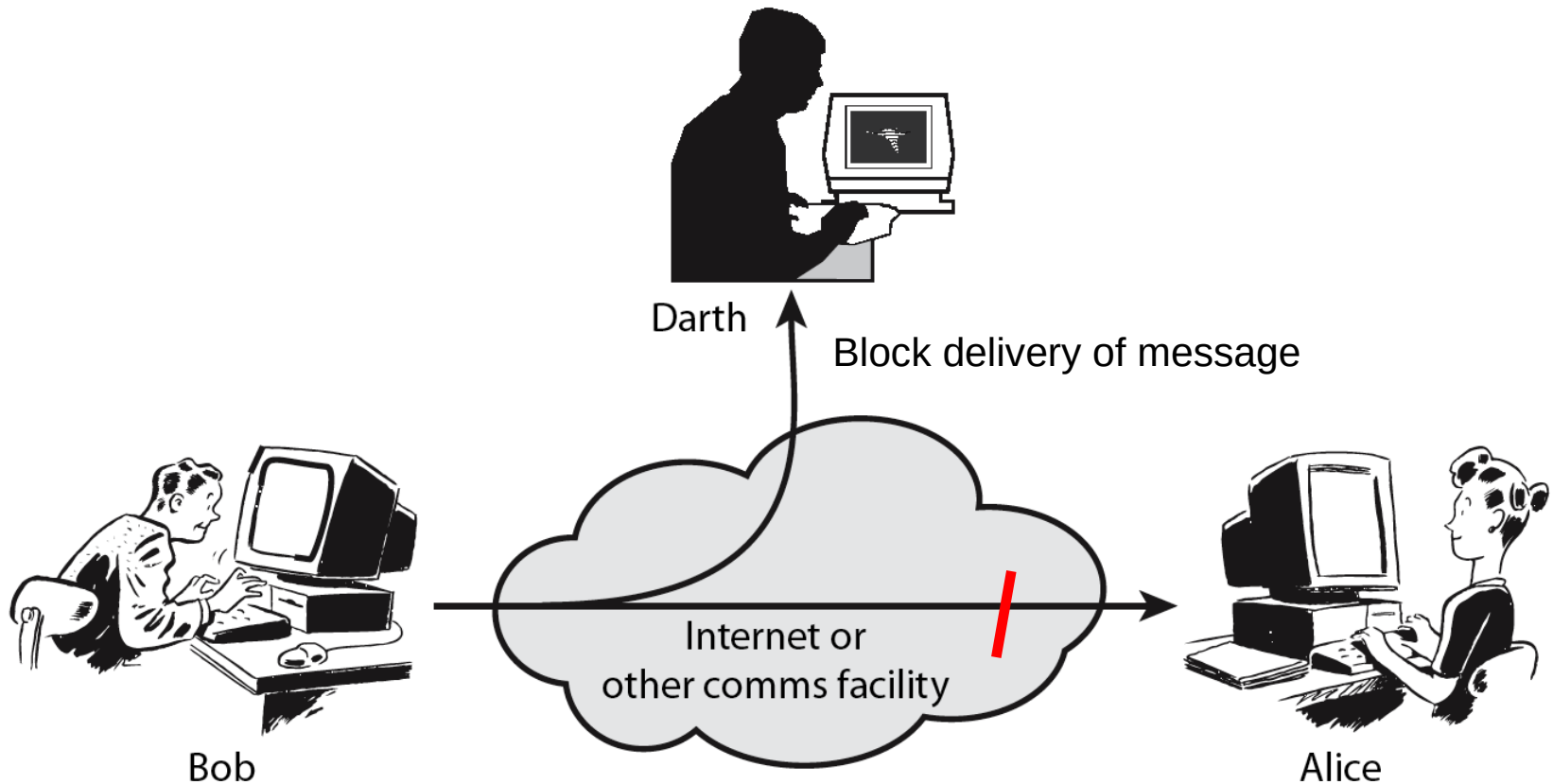
Passive Attack - Interception



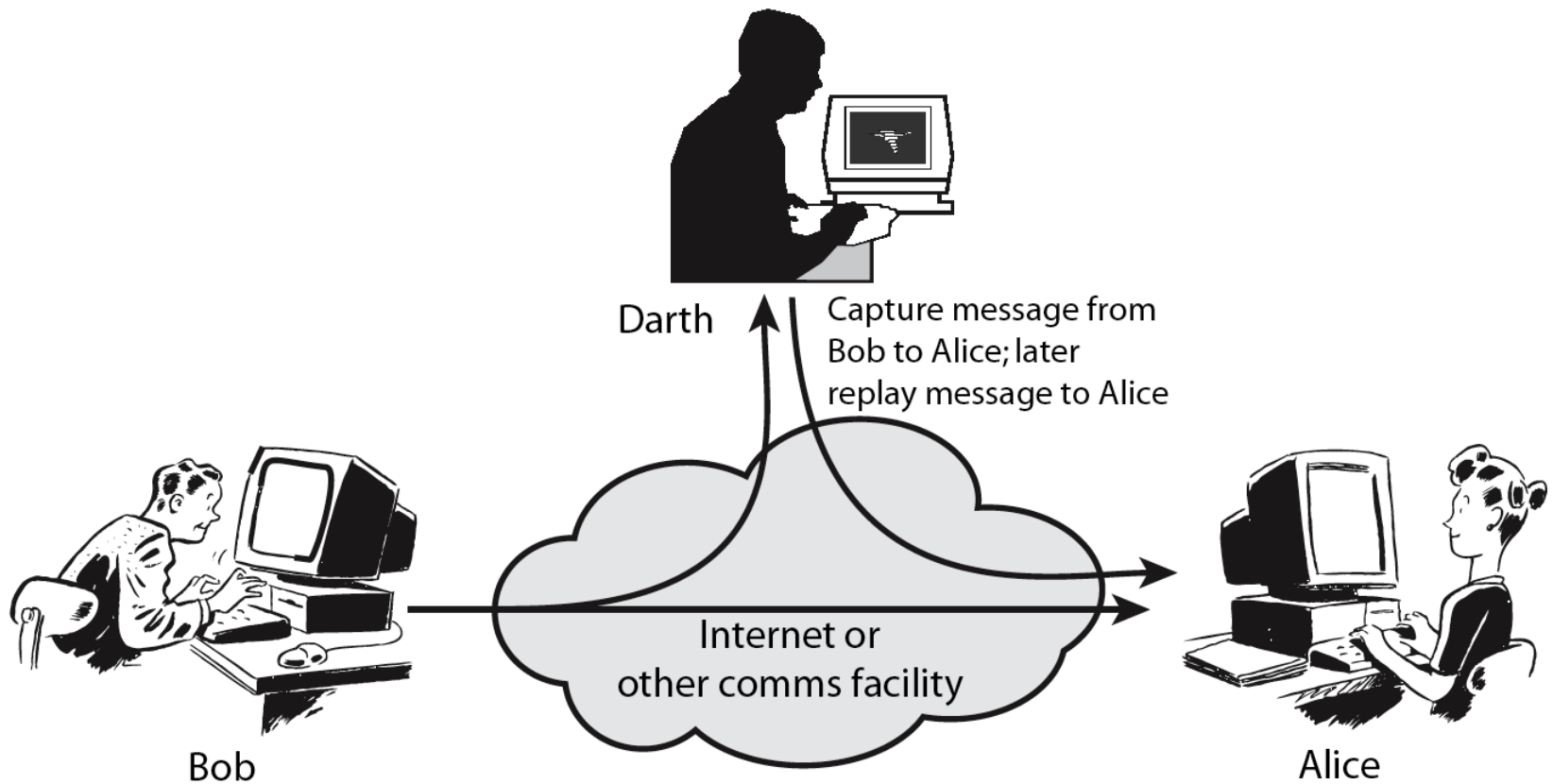
Passive Attack: Traffic Analysis



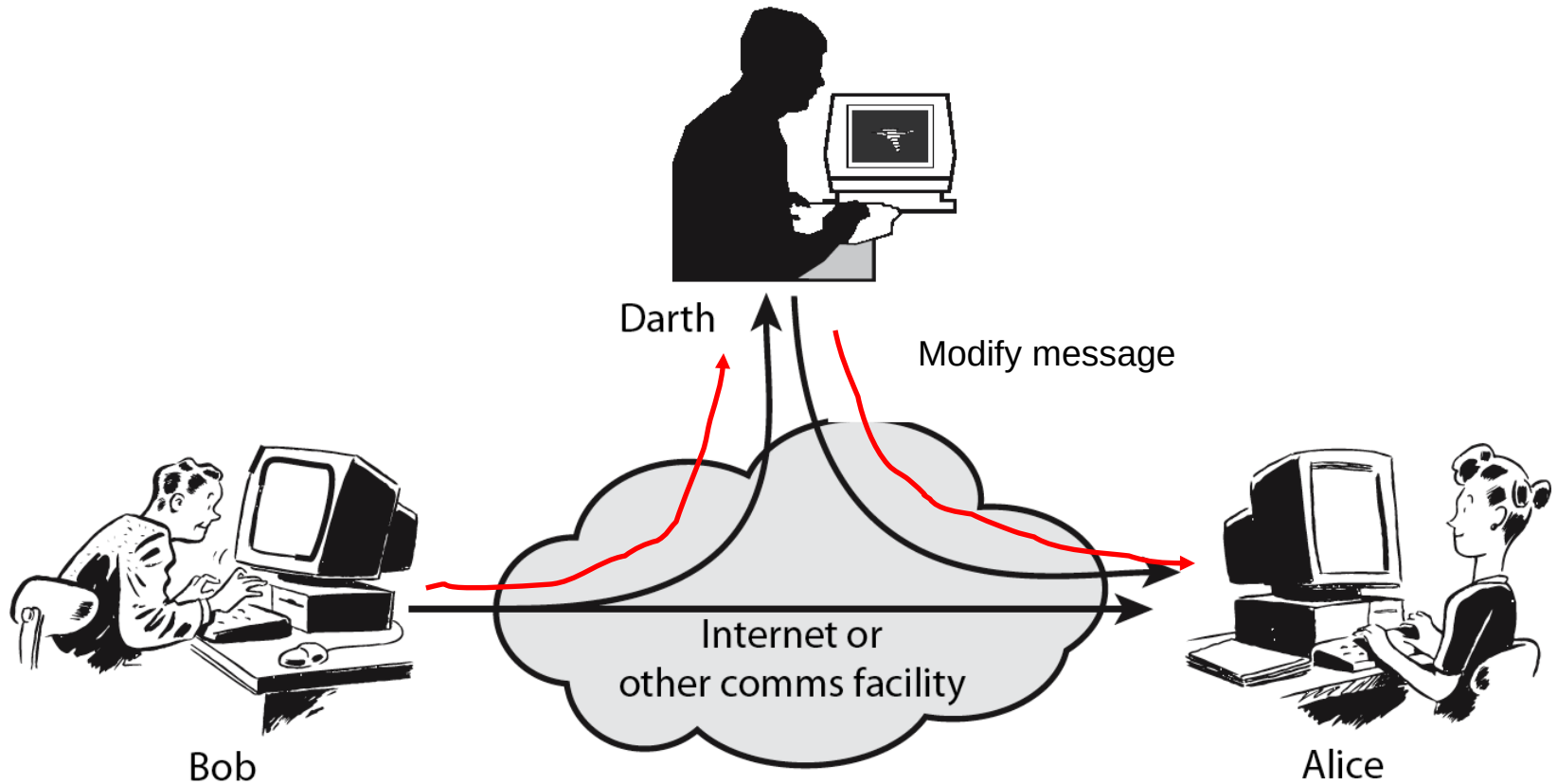
Active Attack: Interruption



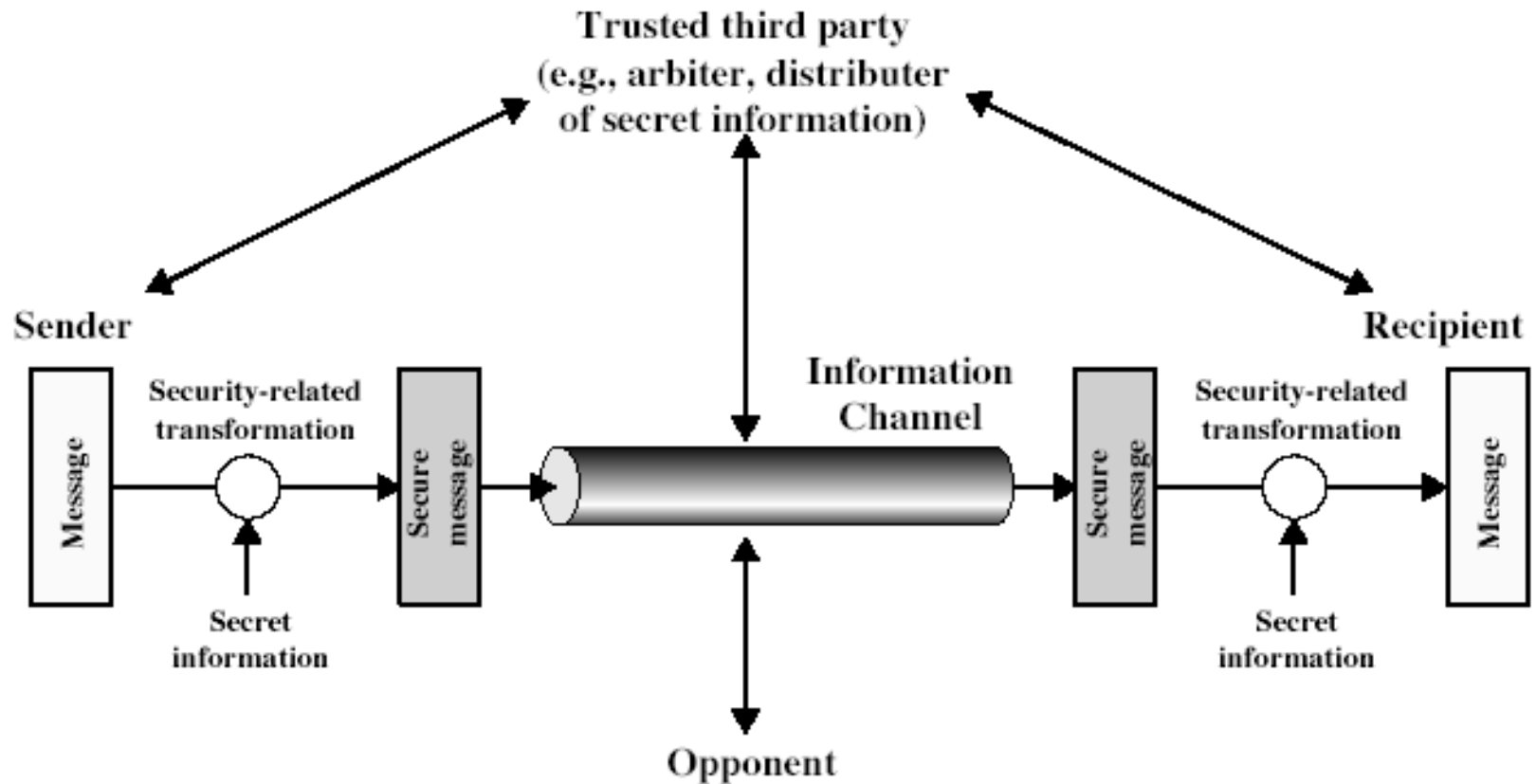
Active Attack: Replay



Active Attack: Modification



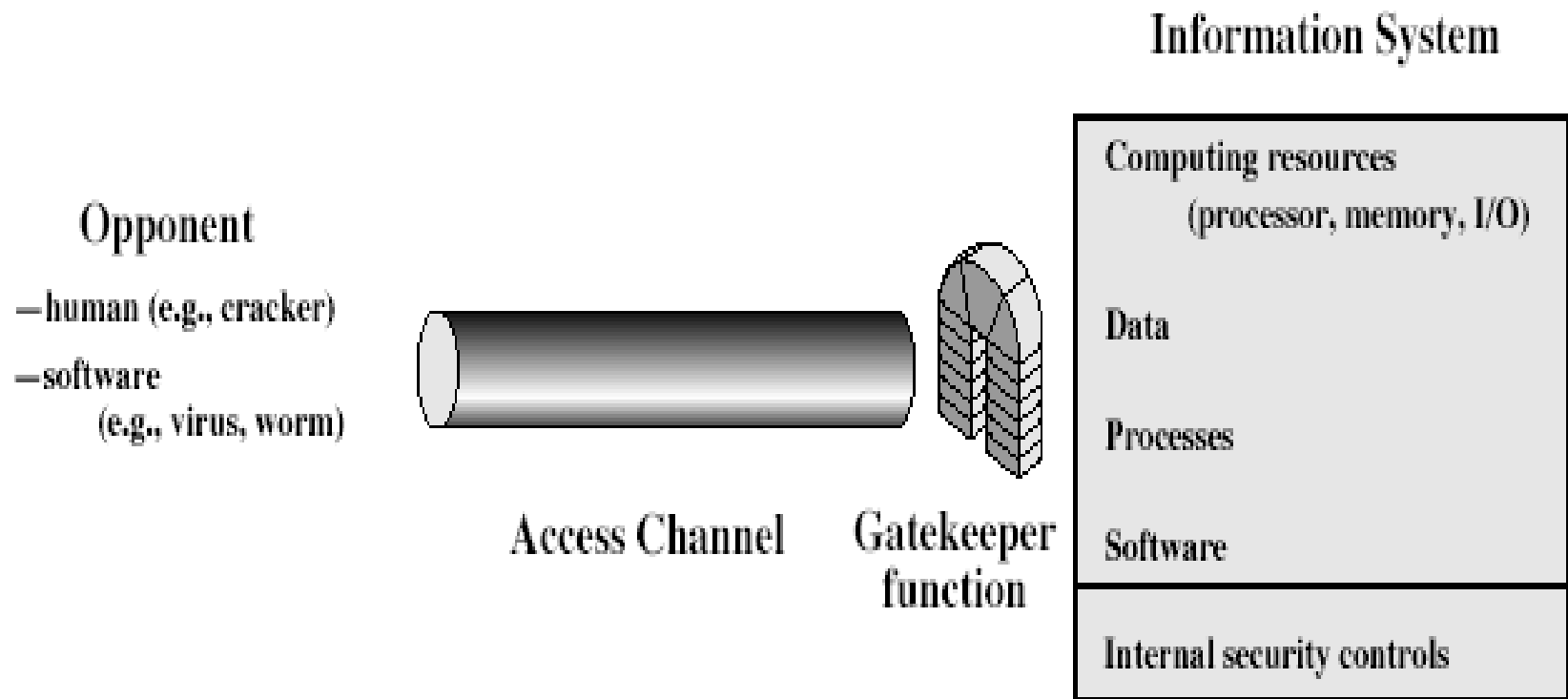
Model for Network Security



Model for Network Security

- using this model requires us to:
 - design a suitable algorithm for the security transformation
 - generate the secret information (keys) used by the algorithm
 - develop methods to distribute and share the secret information
 - specify a protocol enabling the principals to use the transformation and secret information for a security service

Model for Network Access Security



Model for Network Access Security

- using this model requires us to:
 - select appropriate gatekeeper functions to identify users
 - implement security controls to ensure only authorised users access designated information or resources
- trusted computer systems can be used to implement this model

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